

ECO1s002 - Part II - Answers

1 Short questions (6 points)

Maurin and Moschion (2009) study the impact of the labor force participation of close-by mothers on mothers' own labor force participation.

1. Using the social multiplier model, explain why one may expect a positive impact, even if having more inactive mothers around may lower individual utility. (2 points)

Taking care of one's kids may be more pleasant if other people around you are also taking care of their kids. For example, time spent at the playground is more enjoyable if there are other people one can chat with. In the context of the social multiplier, this may be modelled as follows: the utility function of mothers depends on two goods, a numeraire consumption good y and time spent with their children x , with a unit cost p (that could reflect the energy loss due to taking care of the kids, expressed in a monetary equivalent). Around them, the average time spent by mothers with their kids is $S = \bar{x}$, that everyone takes as given. Mothers maximize the utility function $U = U(x, y, S)$ subject to $I = px + y$, with I a measure of their disposable income. This leads to the standard condition:

$$\frac{U_x}{U_y} = p \iff U_x = pU_y$$

with $U_x = \partial U(\cdot)/\partial x$ and $U_y = \partial U(\cdot)/\partial y$ at equilibrium.

Furthermore, one can prove that:

$$\frac{dx^*}{dS} > 0 \iff U_{xS} > pU_{yS}$$

where x^* is the equilibrium value of time spent with kids and $U_{xS} = \partial U_x / \partial S$. What matters is how social capital S changes the relative value of other economic choices. If having more inactive mothers nearby increases the value of a marginal increase in own inactivity more than the value of a marginal increase in the consumption of other goods, the impact of S on x^* will be positive.

Note that this result does not require to assume that $U_S > 0$: it is quite possible that $U_S < 0$ if having more inactive mothers in the neighborhood makes living there less attractive (for example, see the Becker segregation model in Lecture 2, with inactive mothers as type-L residents).

2. Explain why the simple correlation between the average labor force participation of close-by mothers and mother's own probability to participate in the labor force may not be informative. (2 points)

A positive correlation does not necessarily indicate a causal relationship in this case because other factors may jointly influence both individual participation and average neighbor participation, in particular through the determinants of location choice. This issue of endogenous *spatial sorting* may be related to observable variables, such as housing prices (inactive mothers live in poorer households and may only be able to afford to live in cheap neighborhoods). In this case, controlling for local price levels, the correlation measured in simple regressions would still be informative. However, there are

reasons to believe that some of the neighborhood characteristics driving location decisions may not be observed to the econometrician: for example, there may be very local amenities (a new coffee shop, for example) that are enjoyed by inactive mothers. Even more importantly, the social multiplier effect may impact location decisions, triggering a reverse causality problem and the clustering of inactive mothers in some neighborhoods. In this last case, the correlation could be interpreted as reflecting that the presence of other inactive mothers makes the neighborhood more attractive for inactive mothers, but it would not demonstrate that mothers *choose/prefer* inactivity when there are other inactive mothers nearby.

3. What is the empirical strategy used by the authors to mitigate the previous problem? Does it clear all the concerns? (2 points)

In order to assess the causal impact of the labor force participation of close-by mothers on mothers' own labor force participation, Maurin and Moschion (2009) resort to an instrumental variable strategy. The idea is the following: if there is a variable that will only influence the outcome variable through its impact on the explanatory variable, one can use this exogenous variation in the explanatory variable to purge the correlation from the endogeneity problem. The main instrument used by the authors is the gender-mix of the first two kids of nearby mothers, because having two first kids of the same gender dramatically increases the probability that the mother will have a third child (maybe because of a widespread preference for diversity) and drop from the labor force as a result. In addition, the gender mix of the first two kids of nearby mothers is not correlated with individual labor force participation.

Based on this methods, the authors report a 60% elasticity of individual labor force participation to average labor force participation of nearby mothers. However, the *interpretation* of this elasticity in terms of preferences/choices is not immune to the possibility of other causal mechanisms, in particular that having more inactive neighbors may have an impact on individual employability, directly (through referral effects, for instance) or indirectly (through peer effects on individual productivity).

2 Problem: neighborhood tipping (9 points)

We consider one neighborhood, with no capacity constraint. Potential residents are perfectly mobile and they are either "Black" (B) or "White" (W). There are 61 Whites $i \in [0, 60]$ and 41 Blacks $j \in [0, 40]$. Individuals' utility is based on the maximum ratio of the number of neighbors from the out-group to the number of residents from the in-group they tolerate. Individuals are heterogeneous with respect to this tolerance level: the tolerance ratio is uniformly distributed and defined by $T_i^W = 1 - i/60$ for Whites and $T_j^B = 5 - j/8$ for Blacks. The least tolerant individuals always leave first and there is no anticipatory behavior.

1. Who is more tolerant in this framework? Explain with examples. (2 points)

The more tolerant individuals are those with lower values of i or j : for example, individual $i = 0$ is fine living in a neighborhood where there are as many Whites as Blacks, individual $i = 30$ is fine living in a neighborhood where there are twice as many Whites as Blacks, while individual $i = 60$ is only fine living in a neighborhood all residents are White. Tolerance is uniformly distributed between 0 and 1 for Whites, and between 0 and 5 for Blacks so Blacks are more tolerant as a group. For example, 32 Blacks are more tolerant than any White.

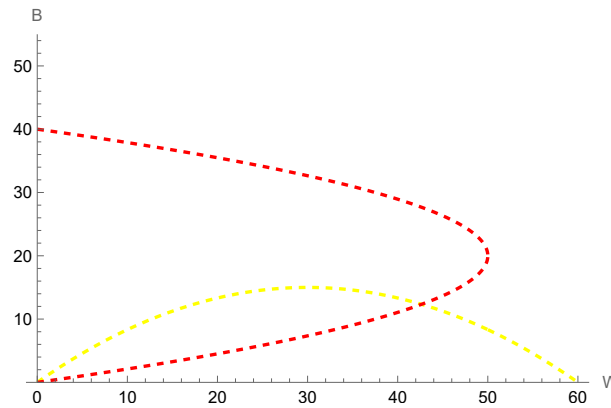
2. Justify the construction of the response functions that solve the location problem. (1 point)

By definition of the tolerance ratios, we have $T^B = W/B$ and $T^W = B/W$. Given the uniform distribution of tolerance ratios, the number of members of a given group is equal to the counter i or j , so we may write $T^B = (5 - B/8)$ and $T^W = (1 - W/60)$. The resulting response functions are therefore:

$$B(W) = W(1 - W/60)$$

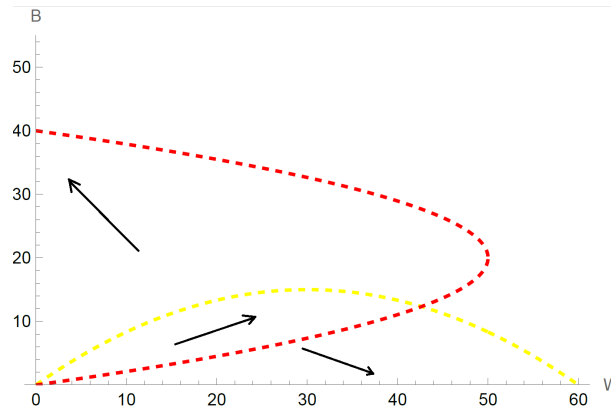
$$W(B) = B(5 - B/8)$$

3. Draw the response functions in the (W, B) plane and use your graph to find numerical approximations of the solutions to the system of response functions. (1 point)



By looking at the graph, one can find approximate values of the solutions, corresponding to points $(0, 0)$ and approximately $(43, 12)$.

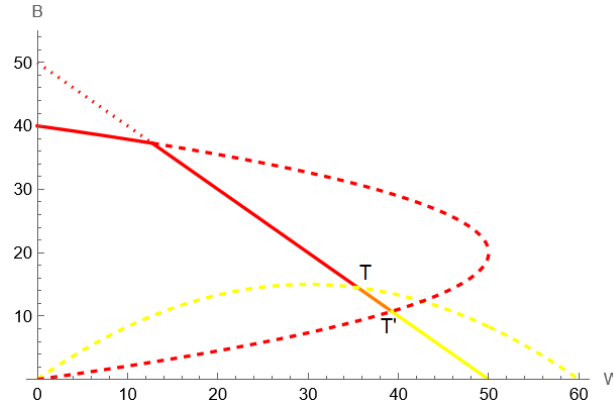
4. Use the graph to describe the possible population dynamics. (2 points)



The dynamics of the model are summarized by the arrows on the graph. The final equilibrium outcome depends on the initial inflows of population. There are two stable equilibria, reflecting full segregation: points $(60, 0)$ and $(0, 40)$.

If initial inflows are strongly unbalanced in favor of Whites, no Black will ever move in and the population will converge towards the $(60, 0)$ equilibrium. Conversely, if initial inflows are strongly unbalanced in favor of Blacks, no White will ever move in and the population will converge towards the $(0, 40)$ equilibrium. Conversely, if initial inflows are somewhat balanced between the two groups, the population will grow for both groups. However, no interior equilibrium is stable here. Even if the population manages to reach the interior equilibrium $(43, 12)$, then more Blacks will still be willing to come in, and they will trigger the departure of Whites, leading, once again, to the $(0, 40)$ equilibrium.

5. Assume now that there is a fixed supply of 50 houses in the neighborhood. Draw the supply constraint on the graph and with the help of the graph, explain how this augmented model may be used to explain neighborhood tipping. (3 points)



The supply constraint is represented by the straight line of equation $B+W=50$. In this model, allocations above this line are not feasible. All the allocations represented by the orange segment are stable because even if there are more people willing to enter the neighborhood, they are not able to do so unless someone leaves and vacates a house.

However, if demographic changes are such that leavers on this segment are more often replaced by Blacks, the allocation will progressively move towards point T, which is a tipping point: beyond it, no White is willing to live in the neighborhood, which becomes fully segregated at point (0,40), and given there are more houses than Blacks in the economy, 10 houses are left vacant. Note that, although historically less relevant, the model allows for the opposite tipping process involving point T', beyond which all Blacks leave and the neighborhood converges toward point (0,50).

3 Data work (5 points)

We use a French dataset on couples i observed in $year_i \in \{1996, 2002, 2006, 2013\}$ and estimate by OLS the following equation:

$$EDUC_i^p = \beta_0 + \beta_1 Age_i^r + \sum_{k \in S} \gamma_k (EDUC_i^r \times \mathbf{1}_{size_i=k}) + \eta \times year_i + \varepsilon_i$$

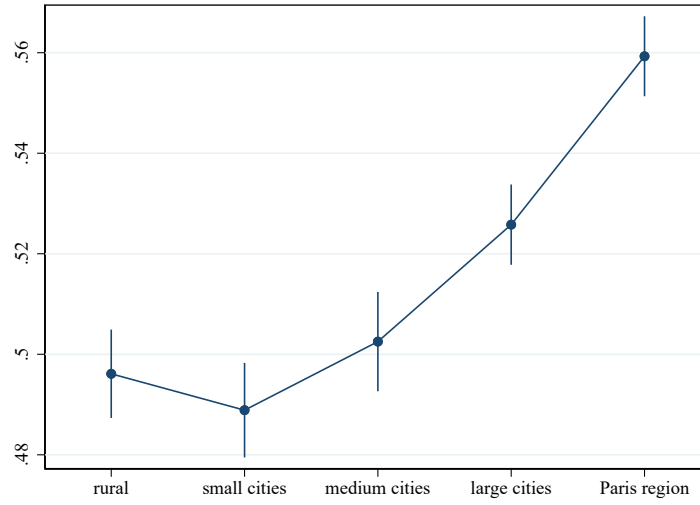
where r stands for “respondent”, p stands for “partner”, $EDUC$ is the number of years of education, Age is the age in years and $size$ is the size of the city where the couple lives, split in a set of five classes S .

1. Why do we control for Age^r and $year$? Can you guess the sign of β_1 and η ? (2 points)

Respondent's age may have an effect on partner's education through its impact on partner's age. This will be the case if the level of education changes with age, and if both partners in a couple have a similar age. Similarly, if the average level of education changes across time, we want to control for a time trend in order to measure average correlations over the 1996-2013 period. Given recent trends in France (as in many other countries), we can expect older respondents to be less educated and the general level of education to be increasing over time, so that β_1 should be negative and η should be positive.

2. Figure 1 shows the OLS estimates of $\{\gamma_k\}$, with their associated 95% confidence interval. Describe in words what this represents and relate your findings to Becker's model of marriage. (3 points)

Figure 1: Regression results



The coefficients γ_k show how the correlation between the level of education of the respondent and that of his or her partner changes with city size. Small or medium cities do not show a larger correlation than rural areas: the correlation is roughly equal to 50% (there are some differences, but they are not statistically significant, as shown by the overlap of the confidence intervals). This means that, after controlling for respondent's age and the year of the survey, respondents with one additional year of education had, on average over the period, a partner with 0.5 years of education (relative to sample mean). Conversely, the correlation is 2-3p.p. higher in large cities, and 6p.p. higher in the Paris area.

According to Becker's model of marriage, positive assortative mating will occur under supermodularity of the marriage output function, or if members of the couple split their joint output based on some predetermined sharing rule. Large cities may be more efficient marriage markets because their larger population allows for more diversity (individual indices are more evenly distributed, so that everyone can optimize). Conversely, it may also be the case that large cities offer more possibilities for a supermodular joint production function, for example, with childcare services, or better career opportunities for both members of the couple.